

ActInf Livestream #025.1 ~ "The Computational Boundary of a Self"

Livestream | 1:54:10

hello everyone

welcome to acton flab live stream number

25.1

today is july 6th 2021 and we're going

to be talking about the paper

the computational boundary of a self

developmental bioelectricity

drives multicellularity and scale-free

cognition

we are a participatory and online lab

that is communicating

learning and practicing applied active

inference you can find us at the links

here on this page

this is a recorded and archived live

stream so please provide us with

feedback

so that we can improve our work all

backgrounds and perspectives

are welcome here we will be following

good video etiquette for

live streams here at the short link you

will see all of the live streams and

different series that we've done leading

up to this in the communications unit of

the active inference lab

and today we're going to be jumping into

the first half of

a discussion of the paper uh in july and

on the 13th

we will have 25.2 which is also on this

paper and hopefully with

the author joining today

in act in flab live stream number 25.1

we're going to be learning and

discussing this paper

uh so i will um pass it off i'm blue and

i am an independent research consultant

based out of new mexico and

i will pass it to steven for a quick

round of introductions

thanks blue hi so this is stephen here

i'm based in toronto

um i'm doing a practice-based phd and uh

i'm interested in how active inference

can help unify

lots of threads that have been very

confusing to me over the last few years

and if it's okay i'm gonna throw this

over to dave

douglas yes hi i'm in the philippines up

in the mountains uh the rainy season has

begun

and we just survived the fourth of july

so all is good

thanks back to dan i guess thanks uh i'm

daniel

i'm in davis and yep looking forward to

this

point one discussion and also nice job

with the broadcasting and facilitating

blue

thanks so we can just start off with

just some questions about

what you liked about the paper or what

was something you remembered or

something that you're curious about

well i am embarrassed to say i didn't

read it at

all and i was you know i i started

looking over it as oh my goodness

we're going to talk about wilhelm reich

we're going to talk about orgones and

orgasms and cloud busters and
and living flying saucers and i guess
we're not going to get any of that so
i don't know
cool well the aims and claims of the
paper um
so here the paper the author says the
aims were to define
individuals themselves in a way that
facilitates taxonomy comparison and
communication
with evolved created biological
artificial and
exobiological agents and he proposes a
fundamental definition of an individual
based on the ability to pursue
goals at an appropriate level of scale
and organization
and suggests a formalism for defining
and comparing the cognitive capacities
of highly diverse types of agents
what does anybody think about that
yeah i i think that the multi-scale
part of this is very very interesting
very exciting because
this is one of the big offerings that
active influence has
and it's been offered in a a lot of
simulations in silico context but
to have it actually described from a
real biological
kind of cellular perspective
and is for me really really useful
because i'm kind of looking partly at
how the heuristics of all this are as
much as the
you know the mathematics of it so yeah i
think that's really really interesting

cool daniel i liked the part about taxonomy because that kind of speaks to this evolutionary worldview and it always seemed interesting to me studying collective behavior like in the ants that there were really precise ways of talking about comparing some physical properties in biosystems like how hard is the material or its density and then there was less of a way to talk about similarities in cognitive processes whether on an organism or how they interacted with their niche especially so just the idea that there could be a new sort of tool in the toolbox to help us compare different cognitive systems that might have hybrid aspects like computer and human or agent and niche those are kind of cool areas to explore so i think that's a big topic definitely like the other maybe um like comparison of like cognitive capacities isn't really related to cognition but it's integrated information theory right so you know when you can compare like the consciousness of a slime mold versus the consciousness of a human this kind of stands uh alongside of that

in in terms of
you know being able to to maybe not
quantitatively yet but
hopefully um you know people will
run with this and be able to apply some
quantitative methods to that
uh cool so he the author
also proposes a plausible naturalistic
framework for the evolutionary
scale-up of cognition from the earliest
origins of life
and hypothesizes about the forces that
drove it and the major transitions along
the continuum and the goal is to show
how complex agency and goal directedness
evolves naturally
from ancient mechanisms
uh so some questions how do multiple
nested scales of individuality work
uh that's super interesting to me let's
see
um daniel yeah
well i wrote the question so maybe i'll
ask or bring up why i thought it was
interesting
uh is there like a total value
of cognition in the full full stack here
the full cone
and then there's going to be certain
levels that are more relevant than
others
because they're a total amount of
individuality
or are there just certain ways
that like the the cones for example
don't overlap
like there's certain spaces almost that
are implied that are

more than the sum of the parts and so
how's
how are we gonna even just first draw it
out
think about it
actually one thing that comes to mind
when you talk about the sum of the parts
i think
what i i'm interested by when he talks
about a cell
and then he talks about these groups of
cells and how they are sort of joined
and there's this kind of bioelectric
dynamic that sort of helps them
communicate i think that's a really
um making a bit of a jump here but i
think it's interesting
to the difference between people
communicating in a community
context where there's this kind of
resonance going on between people
and what it's like when there's just
separate entities
sending messages to each other so to
speak um
there's something about this um ability
to
form some sort of bioelectric
um dynamic which i think is
in itself quite interesting as a
heuristic
yeah for sure daniel agreed we're gonna
bring it go to bioelectricity soon
because i think
with um the author visiting next week
it'll be good to
get some of these notions that we're
having about where

bioelectricity plays a role like those
are some of the
big theoretical questions like what is
individuality across
levels is it informational is it
evolutionary
is it functional all these questions
about like holism and reductionism more
than the sum of the parts
what is collective behavior what is
emergence there's that whole
body of thinking and then
the piece that's tying it all together
is the bioelectric
and there's a few reasons why in the
paper that
is being proposed as important like the
primacy or the primitiveness
um and also just the um signaling
dynamics and the
what it appears to allow cells to be
able to do we can explore that later
so agree like how is bioelectricity
supporting this broader story about
multi-scale systems
which doesn't just by drawing it
out this way doesn't imply
bioelectricity
so where is bioelectricity coming into
the picture
we'll keep thinking about that
yeah absolutely i'll be we'll talk about
that later i think and it's it's
something that i don't think
and correct me if i'm wrong other papers
on active inference
they have not tended to come at that
aspect it's it's they've tended to come

at it from a
in silico kind of statistical
state-based
process so this this does give something
that's a bit more um
organic i suppose potentially so i think
that
there's maybe there's much more that can
be drawn from that going forward so
thanks
yeah i think the um you know people have
done
more organic uh stuff but it's always
like even related to bioelectricity but
i think
it's always related to neural
bioelectricity as opposed to non-neural
bioelectricity which is what kind of
drives this paper forward and the you
know of course there's a relationship
between bioelectricity and the
homeostatic drive
which the author talks about as driving
the major evolutionary transitions
and you know probably much of what we do
much of how organisms think and behave
is driven by you know that that drive
for uh homeostasis
so uh more questions what do we
learn or enable when we understand
multi-scale individuality
daniel maybe one way to
frame what we could learn or enable here
would be like in a read
and write with respect to
designing new kinds of multi-skill
systems that's like the right
and then to look around and discover the

kinds of multi-scale systems that we're
already interested in with like reading
so it's kind of there's an interpretive
ability when we can look across
multi-skill systems but then also
within a virtual machine or within some
sort of
physical system there's ways to just
design systems that have certain kinds
of relationships between the parts and
the holes like
a governance system where such and such
a distribution
of of uh influence exists or something
like that
stephen stephen
one area that's been of interest and i
think this can relate to
is in an area called mental space
psychology they've been looking at
peripersonal
neurons the mirror neurons as people
talk about them as well
and how they resonate and i think that
um
this that ability to get that dynamic
relationship between people or with
objects
in our shared model and the ability for
these
to see that there's some sort of
resonance in terms of how things relate
to others in space
that that seems to tie into what you
were saying about the bioelectrics
of um neurons tightenings
have been sort of related to this and
ties into that ability

for maybe us to re or reconnect some of these

um inference processes

across distance you know because

obviously the cells they're touching

each other

there's a plausible root but how do we

maybe get some of these

resonances between cells which are not

touching each other

or even organisms which are not touching

each other or maybe they need to touch

and that gets us into intimacy and stuff

but

yeah good point that like brings us so i

just pulled up the slide

um talking about gap junctions which

these are like

you know it's their little holes that

enable

you know those proximal cells to

share information essentially um about

their environment or

their intracellular um

like state of existence and that's a

cool

um way for you know that it facilitates

the

the informational spread um and

we see that in the figure also let's see

let's come to that figure

um so this one yeah

so that gives you like you know if if

there was a boundary like a solid

boundary

or maybe even a lipid bilayer that

wasn't didn't have a little hole in it

connecting cells then there would just

be the individual spatial
perception that's shown here in figure
three but
because of the gap junctions um you know
a cell
communicates to all the proximal cells
and
this expands the spatial perception and
the memory and the
anticipation and so it's like this group
integration here
uh dave
yeah going even farther into field
effects
um i heard a presentation by some
students in china
recently who were tracking down
non-uh
tracking down electrical effects in
mammalian nervous systems that
didn't that were not confined to
local chemical and electrical effects so
they didn't even need to use the pores
they said this is not dominant
but it's there and under certain
circumstance it does become pretty
important like in um
long-term potentiation they're saying
that's part of what's going on that you
get these standing waves
and they're then they're uh conjecturing
that
that part of the classic brain waves are
literally are waves in the brain
rather than a summation of these
local uh synaptic jumps
don't know daniel oh
daniel yeah just one comment on

what dave just said there about these
kind of traveling
fields that is really related to
kristin and collaborators work with spm
which was basically a toolkit for
looking at dynamical systems
and then integrating imaging based
modalities for like neuroimaging and
with the field-based measurements like
eeg
so kind of taking a certain anatomical
shape into account
when you are integrating these different
projections or different
modeling ways of kind of
two different uh perspectives on the
same underlying system so that's kind of
an interesting
link that you just
made there yeah it's hard to under it
it's hard to overestimate the importance
of
professor fristen's enormous amount of
hands-on
work with interpreting brain
direct physical brain data and how
important that is in just implicitly
shaping how he
uh approaches uh these system
activities and system effects yeah i
keep
yeah i keep slapping myself wait a
minute what does this guy spend the last
40 years
doing you know don't get stuck in the
peculiarities of the
mathematical descriptions he has a real
reason for doing this stuff

that way dania
you go first stephen oh okay yeah sure
well i suppose
being a bit speculative here but with
the brain waves in the way that we've
got this bioelectrics on the cell
i mean different scales of i think this
is an open question but it certainly
seems to have plausible
roots to being thought about is how much
of these
brain waves or this electrical activity
across
regions of the brain analogous or in
some way
similar to the the descriptions at the
cellular
and multi-cellular level so i think
that's a good point that
dave brings up there
thanks daniel yep great point
stephen and that's one of the main
i think arguments or the ways that the
field or understanding of
biology would sort of shift if
bioelectric were given a
larger role there'd be more of a
continuum of
chemical and electrical function between
internal nervous systems and
the epithelial cells which like we
talked about in the dot zero
are closely related through development
and other cell types and seeing that as
more of a unifying framework within an
organism's
cognition as well as in an evolutionary
sense like all these

different affordances that the
bioelectric reconfiguration
allows for which is some of the work
that's cited in the paper and
has continued since then is just
empirical demonstrations that the
bioelectric field can like lead to
different morphologies
and so then there's kind of a way to
move on that phenotype landscape that
doesn't
involve like only the um sequence based
remodeling of gene regulatory networks
like transcription factor binding site
evolution could still be important
and also there could be certain points
where the bioelectricity
enables new uh things to new phenotypes
to change as well
yeah definitely and um maybe a little
bit in this paper but
a lot of mike's work is really great uh
talking about
um or or demonstrating you know the the
role of
bioelectricity in shape and you know
when you think about like
you there the form and the function are
just coupled together
so like the function of
a cell and the shape of a cell and all
that is is really
um interesting and how bioelectricity
is connected to the form and the
um function uh stephen
yeah and i think this also ties or
certain gives reason to look at some
other

people who presented such as the work on
um
quantum contextuality that we had and
that some of the information geometry
work and there's the new paper on um
bayesian mechanics and active inference
this whole idea of
the shape of information which
i mean that is very cutting edge but i
think they
they don't seem incompatible let's put
it like that
i don't want to put my name to how
they're compatible exactly but this
certainly seems to be an awful lot of
things here that would
suggest their plausible connection
definitely uh daniel oh
daniel so one other aspect there's the
extended spatial that's this horizontal
representation where whether it's the
bacteria that are aggregating in the
biofilm
or the development of the embryo
or whatever extended spatial system
maybe even like networks of computers or
something
like that that's extended spatial extent
in the snapshot and then there's also
this
blue going down visualization here like
with memory
fading off into the past and then also
the green
like the anticipation so that's the
current state
anticipating out forward so just kind of
using a uh visualization of kind of

space going out and time as this
axis that goes through the current
moment that
turns out to be something that
visualizes the past present and future
really nicely but then also like we saw
with the light cones
and probably other approaches it kind of
sets up the
integration of memory action in the
moment and
anticipation in a way that's almost like
set up to be addressed
with a commonality rather than a future
oriented framework
that has to bolt on this like memory
module it doesn't really make sense
but this is a more integrated way to
think about the past present and future
nice uh stephen yeah i like thanks
daniel for bringing that up actually
that's really helpful and and i one
thing that i find curious as well is
even though we're talking about cells
we think about how that diagram puts the
future up
and the past down right as we think
about could have done it the other way
but that's naturally how we look up
and the air you know to see things and
things go down
and a bit more in the earths and that's
where our history is so
even though it probably was not
conscious that would have just felt like
the right way to design the diagram you
see how
um you know the way we make our meaning

is
is embodied as well the way we make our
diagrams is
in ways embodied
yeah nice definitely daniel so that
steven that kind of uh whether
something's pointing up or down
it reminds me of in evolutionary biology
where some subfields
would have a tree that branched up so it
looked like a tree
other fields would show it branching
down and then there was like apparently
disciplinary
like debates over which which rotation
of the same information it quote really
was is it a tree or is it you know
something else but then that was related
to a question that
i just had about the visualization
so it's cool that these panels were kind
of
maybe something we can ask about next
week they seem to be made like in
collaboration
with an artist so that sounded like
something kind of interesting
to have in a scientific paper it helps
it be more discussable
sometimes but maybe there's a learning
experience for
the people involved and then also just
like it's
pretty easy to visualize the chemical
part like we have these beautiful you
know walking protein visualizations
kind of like okay proteins kind of look
like that and they bump into each other

but then how will we visualize the role
of the
bioelectric greeting scott nice to have
you
how will we visualize that um
in new ways maybe by integrating the
future and the past into it or i don't
know maybe there's some other way like
the diamonds that we saw
so yeah yeah i think that's that's to be
a really good question to ask actually
because um
yeah i work quite a lot with in graphics
studios where they do
different types of diagrams for
different things and i think
there's a lot of stuff that conventions
that are out there so they must have had
to
also bring in and sort of make it make
sense in the scientific paradigm but i
do like the way that
they've they've used the term perception
there and brought that in for a cell
i think that's um again unusual but
makes sense if we think of perception in
a much more deflationary sense than how
people traditionally think about
perception and active inference as well
as
just some sort of way of knowing um so i
really and again that also starts to
bring in
like cells when we think about
what is the meaning that's not how a
cell is
cells could have a sense of knowing and
there could be a hundred billion cells

knowing which is pretty much what
indigenous knowledge says
about knowing so it's another it ties in
with that regime of knowing which
is often minimized and excluded so i
think this is quite
quite useful nice and just kind of to
talk about like different
renditions of like the light cone i mean
this is a question that i
definitely have for the author uh here
this is like not the traditional way
that i'm used to seeing a light cone so
this in this slide you can kind of see
how it's more of like a conical shape
as opposed to like a diamond um and like
in space and time um and even like so
this is up and down
um but you know there's also this
rendition that i've seen as well that's
left and right
so like from the from the state of right
now to the
future and so it's interesting to think
about
why is the space and time bigger
in the present and going to a more
narrow
point in the future like for me that's
kind of not
how i'm used to conceptualizing um these
diagrams so i think about the space
of right now like i can't do anything
other than what i'm doing right at this
moment
but you know potentially the the as time
goes on my
my state space gets bigger if that makes

sense

potential state space i i really like

that as well because

the thing is it's interesting that cone

is a bit it's very popular in this

strategic foresight field of looking at

futures

and strategic futures work but there is

a slight

like when it's done as a cone going into

the future it's very

visual it's like it's like using the

analogy of a beam of light

effectively again using one modality and

it assumes it's kind of

travels in a straight line and it

projects and it's

whereas when you start to get into this

non-linear swarming

type thing i mean in some ways it's you

just see what rises up in the future you

know i mean it's like

what what structures emerge into the

future um

they don't project into the future in

the way that a lot of strategic

foresight so i think this does

that question about how those two

different models

manifest and get used and how this other

type of model actually while it's not as

easy to intuit

may actually be more more i don't know

if we say accurate but

yeah maybe more accurate is better way

of saying it yeah i

i actually um definitely agree because

you know not to be morbid but we all are

progressing
to the same end right so
and as we age our affordances are less
and less like you know we may we might
not be able to
you know cognate as well or or we might
be less mobile as we as we get older
uh daniel sorry did i take the words out
of your mouth
um how about dave first and then i'll go
okay dave
yeah so is that becoming more focused
into the future does that have anything
to do
with most of us becoming more and more
subject matter experts rather than the
other kind of
expert
daniel nice
question dave so um the the sort of
duality of the time cone and the time
diamond
it just it just suggests that there's
certain things that we want to think
about as being maximized in the present
maybe and certain things that are being
as if they're minimized in the presence
so it's just two different and we'll
figure out maybe
in some ways there's some things that we
know off into the future
and there's other things that are
maximally constrained in the present
but then to connect that to the
stephen's question about um
strategy and um you know ask scott what
he was
uh interested in with uh you know

joining us it made me think about the
width of this cone
going out into the future or the the
length of the one that's more like a
diamond
it's kind of like the horizon which you
need to consider
for de-risking or for strategy and so
that space
just like the butterflies wings that
space expands
super rapidly now you might have a
hypothesis about the space that only one
region is likely to be
um explored or that all of your strategy
would
would you know work across this whole
space but it just sets up
the discussion instead of like time
points and
bifurcating points it's like actually
it's going out really vastly
so there has to be some more adaptive
way of realigning because
the time cone is going to be too large
to search through in the future
even on a short time step so how could
we enact a strategy in the current
moment
given what something what we are
maximizing so that
we could actually have an adaptive
strategy moving forward
nice dave and then stephen
no i just hit the button wrong got it
stephen
yeah i saw scott's got his ice cream
cones there so i'm gonna be curious to

hear his his point but
um tying with what we're saying is also
we have this because swarming dynamics
and non-linear dynamics basically
aren't part of the most systems
approaches
really there's an assumption that it's a
volume of information problem
going from the present to the future and
it's just finding ourselves in all of
that
but um and at times when we've got the
like in modern world we've kind of
thought the world is much more
predictable than it is and maybe it was
when it was in a very constrained
system but now you know we can see that
even in
our own social systems the things can
once they bifurcate that whole
cone thing breaks down because it
suddenly it's
it just um so i think that there's
there's
there's something quite um useful here
actually in rethinking and
using this alternative or maybe
something which is a
other version of this um this this this
version of this cone
um which doesn't rely on ocular
light visual because there's no eyes in
a cell
for instance they're not seen into the
future right there
that's only something that happens at a
quite a aggregated organism
so um this is this is actually probably

has some other spin-offs that's quite
useful
scott this is a
fascinating conversation so by the way
this is a cone that i saw in new york
that it was an ice cream cone it was
very impressive it had nine scoops on it
so that was that's the kind of light
cone i like
the um you know i was thinking back of
that now the knowing
that you were saying before blue and
it's so interesting
so what if we perceived a group as a
bunch of cones
a bunch of simultaneous cones and
because if they have embodiments or
shared meaning
physically or mentally then their cones
have a overlap or a similarity or
shared space and but you can't really
experience another
party's cones or or can you indirectly
um so that notion of knowing that really
hooked me when i just
came into the call that you have these
cones going on at the same time these
experiences
that are through time and maybe
when we read literature what we're doing
is we're experiencing someone else's
cone in the past
in order to try to project it onto
our experience in the future and then
maybe you live by that rhetoric maybe
the rhetoric actually becomes your
future
right so if your mind can have a certain

state that gets acquired by contact with
other states

it makes a conformational change kind of
like prionic

governance we have a conformational
change in a protein when it misfolds
maybe have a conformational change in
your cone

state when it comes in contact with
another cone that has a different
folding

so that the narrative maybe there's a
narrative

element to the contiguousness of the
cones and because they have a temporal
and a physical element to them but you
can only experience them in the present
then the rest of it is just projection

either in the past or the future
and so maybe that's maybe the cone is
entirely rhetorical

and maybe it's convincing yourself and
others

of the state that you have for
existential continuity

and the ways of knowing maybe goes to
that in terms of multiple

cone generators trying to get together
and figure out if their little cones
equal a big cone

it's really just a fascinating way to
visualize the space

there's one other reference i want to
make which is from the museum of
jurassic technology

in california culver city they have been
displayed by a guy named uh theory by
sono bend

s-o-n-n-a-b-e-n-d and it's called
theories uh theories of memory or
theories of forgetting something like
that
and his whole thing was cones
reseeding and approaching light cones
and the way he divides up space and
knowledge
is by taking different arcs and sections
of these cones is really kind of
fascinating stuff
so really nice scott i'm reminded a lot
of
um shanna dobson and like the way that
she
perceives time and like also
in microscopy like you can take a
projection of an image
which is where like all of the stacks
from the angles that you're using in the
light cone come together
and you can see it all at the same like
point or it's like flattened out
and it makes me think about visualizing
time like that um
daniel and then stephen so
thanks scott i totally agree one of the
big questions is about
the intersections of time cones what
kind of happens at that intersection do
we think about them getting
like folded up or having some new shape
that arises
and i think a way that we could have a
simple
hands-on intuition there is like with
the two-dimensional game of life
where there are certain shapes like

gliders that kind of propagate on the
landscape
and then there's certain moving shapes
that
perfectly annihilate each other there's
certain ones that leave only rubble
there are certain ones that replicate
but then there's other ones that you
know pass through each other
depending on some feature interaction of
the shape in the
game and then there's ones that uh
do computation like there's of course
great videos of the game of life being
set up to do
an emulation of a of a computer like the
ones that we're on
but slower um so
that really maybe connects a few pieces
this sort of
um agent-based interaction through time
we get our intuition in this grid zone
like the game of life
and then maybe some of those
interactions are happening in
the natural systems just like some game
theory mechanisms that were in the
matrix
of mathematics sort of also play out in
biological systems really nice daniel
uh stephen yeah this this question of
time
also relates to when we say the now
you know we know from even to say the
participatory sense making that
we're anticipating when something's
going to happen often
what we sense is is that there's a lag

so in a way we're always
a little bit behind what's just happened
yet at the same time at certain times
we're anticipating things
before they happen and maybe before we
realize we're doing the anticipating
and then we've made the decision before
we knew we made the decision and then
we've done it so
and which is where maybe reason partly
why sports is so satisfying because
we're
we're making it have to be the baseball
bat has to hit the ball
and you're somehow bringing in that you
know there's no you either hit it and it
goes out there
the baseball diamond or it doesn't but
the same time
when you made the the the cognitive
process
happened earlier the realization of what
you actually hit it
might be later than the now that you
think you're in
so there's this interesting window there
of of what is now
or can you even find the now um as i
think shanna argued in
in her math stream daniel
so stephen with a sports example and
that kind of relates us back to our uh
the skilled
performance discussions that we had
previously so that reminds me of
a bf skinner book beyond freedom and
dignity
where he's like if you imagine a

situation where somebody is like
unconscious or um they're forced to do
something you don't give them that much
agency or autonomy
but then in the uh situations that look
unprompted like somebody walking up on
stage then playing like a
violin performance which is still a
mechanical performance
we regard that highly and we
shower that with the um sort of
aura of agency so he was kind of
questioning that from his own
behaviorist perspective and it's just
kind of cool to think about how
active inference and multi-scale systems
those same questions come up
like there are cases where things seem
really constrained as far as actions are
concerned
and then there's other times where
behavior just seems like come out of
nowhere
from by like a bird song or something
like that so
having a framework for behavior that can
connect
the thoughts across those isolated
occurrences
is really promising
nice and yeah we've talked about
planning as inference and really
the author brings up here um planning in
relation to
the homeostatic space right so so
really this might be the first like like
how planning came to be
is like planning for some potential

state that is not my ideal homeostatic

space steven

yeah both a good point yeah because if

it's already my ideal homeostatic space

why use all that energy valuable energy

to try and predict it when it's already

being done quite well and like

daniel was saying there about skillful

performance i mean

once you get into this groove and

someone's in the zone

you know more and more of the processes

have probably been taken care of by

morphological computing which is

basically the cellular processes and

then when it's at these other temporal

depths

i i'm predicting and just getting right

the emphasis on the third note

before the crescendo or whatever it is

you know it may be that in some ways

when you're in the flow

which is that term when people are kind

of immersed in an experience

you know are you in some ways less

conscious of

more rapid dynamics and

uh if without getting into a debate

about consciousness i think we've got

enough on the table but

um but just just that question about um

yes

it is it is encompassing many more areas

of what we've talked about than that

first

had been apparent

nice and this touches on memory also so

i flipped over to that slide

scott you know it's interesting that
notion just a minute ago about being in
an optimal
or zone space um i heard somebody
quote somebody a few years ago was
talking about different personalities
where some people
like the social equivalent of classical
music and some people like the social
equivalent of jazz
and so the people like the social
equivalent of classical music want to
hang out with the same people
all the time and then oh bill how's
bill's drinking problem going oh how's
mary's
you know uh kids doing you know it's the
same story just incremental changes
kind of like if you hear beethoven's
fifth and it's like
you're like whoa that's slightly off
because your expectation is different
whereas jazz you don't know what it's
gonna be
boop just all different things right so
it's kind of interesting what your
expectations are and what that
optimization
is jazz is a different kind of
optimization right
and so it made me think about the
anterior
cingulate nucleus my understanding is
that's the part of the brain that if
you're
eating at a ripe berry bush and enjoying
ripe
berries the anterior cingulate nucleus

if it fires off
will make you think huh i wonder if
there's a riper berry
on the next bush and so it's the and and
i
i call it the commitment area where men
can't commit to a relationship i believe
it's probably the anterior cingulate
nucleus they're looking for a riper
berry so to speak
so so that i wonder
if there's an interesting rand i'll use
a notion of random number generation
here but
it's probably not that but if there's a
randomness factor that is
in your optimum so is your state when
it's optimized
what is there something in the model
itself that can throw randomness in
there besides the externality
so that and then you become like a
novelty seeking person who has a highly
active anterior cingulate nucleus
is that an optimization which is
superior for instance if you have a
changing highly
rapidly changing environment things like
that just a couple of thoughts on
optimization and what the zone might
look like in different settings for
different
embodiments yeah that reminds me like
i mean it's like a life plan or like an
organismal life plan like some
organisms like have to be in their niche
environment like you can never
take them out but then like some

organisms like prepare for maximum

randomness like i think about a

tardigrade

you know like you can do anything to

them like you can dehydrate them and

send them to the moon and

have them come back and all these things

but but they are they're

you know built they're well suited for

maximum randomness

daniel i liked scott's

point there about the genre of

optimization um

i'm i'm personally always surprised

listening to both jazz and

classical it's just that it's a

different genre and so

a musician in either or both of those

areas though would have like a deep

enough generative model

to know when one thing was off by even a

tiny bit even in a song that might sound

like

um discordant to somebody from

because the improvisation tradition of

one genre like might be

more uh i don't know non-linear in some

way

but definitely this idea of like having

interpretable

or experienceable ways of interacting

with

different optimization processes

different distributed processes

are there more jazz like or more

classical like

uh collective behaviors and then also

this broader

question that is returned to centrally
which is like that cognition
is related to individuality that's
really uh relevant for talking about
where is cognition offloaded or how does
cognition offload into the environment
through development like through
modification of the niche
and then also through evolution where
also the niche can be modified in the
long term but
evolution can select for like
canalization so stabilization
of things that were accessed plastically
in one generation
nice stephen and then scott
yeah i mean falling in on what you're
saying about the environment and and the
jazz and classical um
i used to live in prague for for a year
or so and and what was interesting there
so you've got a lot of classical music
venues and operas and you've got
lots of jazz as well but they're very
different spaces as well right you've
got the basement
jazz bar in some cellar and you've got
so there's a question as well about
yeah once when you start going to one or
the other you probably will keep going
to one or the other for a while
if that makes sense you sort of even
though so beyond the music once you yeah
there is a kind of um
a dynamic because those those two are
fairly clear it's
relatively extremes i suppose even
though they're both what you would call

high art

um in some ways as well so you've got

that interesting

thing about how you bring in the

environment and that's one thing active

inference does which

many forms of psychology and that don't

is how much is it about

changing your generative model and how

or how much is it about another model

about

choosing your environment and letting

the environment do a lot of the

uh motivating for you um so that's

interesting i was just also on a call

um about metaphors and andrew austin who

does that work he talks about

if you want to motivate yourself it's

actually easier to change your

environment

it's an externality that can be helpful

if you want to

change your choice making that can be

internal so like don't try and choose

get others to make better choices

make better choices yourself you might

not be able to motivate yourself maybe

find things out there that help motivate

you

so in the same way that you know maybe

that's good if i want to get a more

ordered kind of structured view of the

world for a while maybe i'll start going

to some classical concerts for a while

if i want to release and start sort of

being more free-flowing maybe i'll go

and watch some jazz

um so anyway kind of interesting

thanks scott i love all of you i gotta
say
this is i mean it's been a few weeks
since i've been able to be
on the calls just for different reasons
and this is
beautiful it's such a beautiful thing
okay so
so i was just showing you my tardigrade
banjo ukulele which is what i was
holding up to the screen so that's a
target painted
a couple of things so you know it's
interesting love the last couple of
comments and that notion of
um access to other
right and that kind of recreational and
de-risking access to the other right you
want to know the externality
because you want to render it innocuous
for survival purposes
and then render it delicious for
aesthetic purposes
right and so a couple of books this is
the the past is a foreign country
by lowenthal which is kind of like okay
temporally
the assertion in the book is the past
actually isn't a foreign country they're
the same
it's us right and so the same thing
temporally
oh they're so different oh that's other
right oh we've come beyond this
but that's our own aspiration then
another book
introduction to comparative study of
private law comparative law you got

in law and language i my assertion that
the mind exists in language and the
brain is just an antenna tuned to it
well law is a formalization of language
and understanding so you have again
comparative study there's another book i
have i couldn't put my hands on
it's the cambridge handbook of
adaptation studies in theater
and the articles on how do you adapt
german theater for a greek
audience or whatever right so all these
things
there's a exotic element to it
right and which is both threatening and
reaffirming and delicious
and scary right and so
these isn't it interesting that those
states
are we other than novelty
perceiving either by resistance or
seeking is is that the essence of
consciousness and so if you had the
equivalent of isothermy
where you had it in isotherm you have
temperature equivalence in two systems
and you can't perform work because
there's no temperature differential heat
differentials
well what if we had no meaning
differential
between two systems is that describable
vis-a-vis that variable as one system
and is that something that starts to get
to you sociality
or shared narrative and is that why
that becomes so existential because the
reinforcements of that becomes

the amplifying wave essentially the
amplifying embodiment

becomes so powerful that it overtakes
the system the system has no other
internal state that's as reinforced and
as amplified

so therefore that one becomes
overwhelming and maybe that's a source
of why it's hard to crack misinformation
by feeding fact or something like that
anyway just a couple of random thoughts
there

really nice scott i um took the tony
robbins

test one time like what's your driving
force in life and i'm totally driven by
instability which makes me like a a
desperate novelty

seeking person um and i'm also reminded
in your points about language i've been
thinking a lot about language

because of some projects i'm involved in
and uh i was listening to

josh tanenbaum speak yesterday about
his um like uh program that he developed
it's like a

neural network neural network like a
sleep wake helmholtz machine

that actually like composes language i
mean it doesn't

it does some like phone numbers and it
but it writes code like computer
programs

um and you know it draws symbols and it
does these cool like things

based on a framework that's um you know
kind of analogous to

active inference where the exploration

phase

happens in the sleep phase of the
machine so

exploration is like this recombination
of all of the potential

like factors right so like in the

training phase it's it's trained on you
know constructs the generative model

or it constructs the recognition model

based on all of the training input like

if it's drawings it's going to be all of

like

you know potential shapes and then it
recombines these

while it thinks about recombining them

to kind of become

a field expert it was really an

interesting paper

uh sorry to detour there daniel

you you go for it scott and then i'll

give a thought

uh thanks yeah just that's fantastic and

it makes me think one of the things i've

been fussing with for a while

and you probably heard me say it is that

the mind does not exist

in the brain it exists in language and

culture and the brain is tuned to the

mind and the example i give is my

niece who was adopted from china at five

months old

now has graduated from cornell she's

looks like a chinese person she doesn't

speak chinese she doesn't like chinese

food she doesn't like chinese music

she speaks spanish and english and she's

a central pennsylvania gal

right so her mind was not there's no

feral consciousness
and her mind was developed in central
pennsylvania
so what it makes me think what you were
just saying is so interesting
so maybe the mind where maybe our models
are just so
so much hubris there and we think the
mind precedes communication
and maybe it just doesn't maybe it is no
mind affordance
other than communication and language so
like plato when he said
the reason that authors were the
challenge is you couldn't query the
consciousness of the author
maybe it was an even a deeper comment
about mind that the the queering of
consciousness
is mind itself and so maybe active
inference
is the function of mind it's not a mind
seeking to
to refine its state more but it is the
fact of mind
is the the act of active inference
it just feels that was that was a shift
that i hadn't thought about before
and it gets away from centralizing mind
in
individual brains that then come in
contact to try to fix up their models
and rather putting situating mind in the
edge itself not the node from a graph
theory perspective
that's fascinating stuff uh
daniel and then dave
yep to kind of bring it to active

inference but really just any kind of
dynamical modeling we shouldn't let this
like snapshot

govern the way that the system works
because systems change through time
so they're not just the snapshot picture
maybe the prevalence of photography
makes it easier sometimes to
imagine this sort of snap shotting and
yes

active inference is hopefully providing
a way to just

think about it again whether
instrumentally or more

in this sort of like philosophical or
metaphysical

way which everyone is welcome

to do but the actual uh core

is just going to be leading to many
questions for a long time

dave one of the yeah one of the uh
environments that is extraordinarily
powerful

at putting people in altered states of
consciousness

and especially distorting their time

sense is a device an enter an

entertainment device which was also

a psychological research device called
music color

stephen may have seen references to that

this was developed in the 1950s and

was in some shows and clubs in london in

the late 50s and early 60s

um it's been mentioned a few times

um andrew pickering mentions that a few

other people in passing

um and there was an exhibit that that

was
uh developed by the same engineer
that tried to play in paris but got
coveted out
if none of you have seen that i should i
have most of the material that's ever
been captured as far as i can tell
because it was
a series of talks a series of literally
chats
uh um in the 1980s so i better gather
that up
and um it's all it's all in a single
series on youtube that i have in a
playlist
so let's do that yeah the the guy would
literally his musicians would believe
that five minutes had passed when it was
actually eight hours they had been
playing solos
for 10 hours at night risking a rest for
trespass
when they thought oh no no the guard you
know everybody's still here i've only
been here 10 minutes buying my flute
and people would get so spaced out they
would
drive their cars into the middle of
roundabouts
up on and bang into the posts because it
just didn't occur to them not to do that
or they'd they'd look into the the u
tracks and watch the train come to them
and
if people hadn't grabbed them and pulled
them off what were you doing
what do you mean i was i was looking at
the lights

and no drugs and they had fun
and the bystanders had no idea this was
happening because the
oh these guys are just playing jazz but
no they were
on cloud9 that's cool it reminds me of a
virtual reality daniel
so it's interesting that with
frameworks for communication like active
inference we can think about
syntactic repetition like repeating the
same word
semantic repetition and novelty like
somebody saying the same thing
differently
but also narrative novelty and uh
similarity
or at least consistency coherency and so
the other piece from what scott had said
earlier was
going back to some of our earliest
streams on
narrative and communication as active
inference
there's a ton of ways to think about
that like
is narrative like a summary statistic
of the agent-based generative models
like you got 25 at this parameter and
75 at this parameter and then narrative
is just like a descriptive state
but what happens when narratives or
various representations of them
can actually through mechanisms feedback
onto sensory and therefore internal
states
so how do we think about narrative and
active inference

and narrative and communication as like
higher order
semantic structures that help reduce
uncertainty
allow for efficient information transfer
all the imperatives of
cell to cell communication bioelectric
but it's happening on a spatial and a
temporal scale
that demands auditory transmission but
of course can also be done electrically
really nice scott
this is i'm my again my face hurts from
smiling
so i so i had this book a rhetorical
style
it's called and it has dozens of
rhetorical
flourishes like the use of french
suggests
style and the use of german in words
suggests
engineering excellence and then there's
different placements
i mean it's just all these different
ways to construct the sentence all these
differences
and it's so fascinating that notion that
is that what what's the cellular
equivalent of rhetoric
right is that when a cell wants to
persuade let's say there's a growing tip
of a mesenchyme tissue or some whatever
in a plant or
a human whatever and it's then and
there's a decision to be made based on a
turing wave that's coming in and what
finger orientation you have right and

the
turing wave responsible for finger
growth there's a periodicity to the
starting and stopping of the chemicals
and that's why you have multiple fingers
so what what's the
notion of persuasion
in non-human speech so a peacock's
feathers are persuading the female
of its fitness because the coloration
one assertion is with coloration if it's
a healthy looking color
or the dance is healthily performed it
suggests a healthy individual and that's
persuasion for mating
let's say i'm just asserting possible
rhetoric
and is all communication
rhetorical in that it's made to persuade
and the persuasion is not necessarily in
the
it may be in the self-interest of both
the speaker
and the externality right because
i'm talking about situations where
you're running an internal model
and you're trying to recruit other
internal models which are your
externality but for other people are
there into or other things are there
internal models right
and so if you want to get some
externality that's an
organism to do something consistent with
your model you need to
er entrain them harmonically couple with
them
so is rhetoric what's the equivalent of

rhetoric in
other non-i guess non-language systems i
use the word language
in non-human language systems
really nice um and i'm reminded of uh
well it made me think of how does the
narrative play across
scales really like from the cellular to
the organ level to the organism all
stephen yeah i think that
also bringing that into what daniel was
saying was structure
and narrative so there's um and actually
that i just looking that up that color
you know the color and how that wheel
relates to sound
music um so you've got these questions
of when do structures start to be
present
that gives something else another type
of geometry and where is that
present so it could literally be um
there could be a type of geometric space
in the way the neurons have arranged
themselves and that
could be a certain level at which
they're getting fed in
embodied information about light sources
so it will only really happen when
you've got an
eye that's measuring light data that
then
starts to structure itself in that form
and that could then underneath so and
that could also apply to
there may be a sort of a quality to the
way sound is
based on it being a whole level of you

know hertz

or half levels of hurts and half tones

so there's something around i think

there's an interesting

point around when we have structure

and we then this rhetorical piece may

happen once we can

oh crikey yeah so you got that as well

the hell so you've got you've got you've

got then you've got systems that we can

see now the question with rhetorical

stuff is

is rhetorical stuff the stuff that we

can see as a system

that you can then argue your point you

know some sort of way this and i suppose

at another level the cells could come to

a point where they could

argue a case with each other and so

there's some system in place that sits

on that structure

but the thing that always is in the

background which is

again what i was saying earlier most

people doing systems

don't think about when they talk about

emergence

they is is there some sort of swarming

dynamics happening

now whether this bioelectric stuff

sits and adds a new layer i mean i

noticed on one of the slides there it

talked about kind of like a new layer

of um uh like does this add a new layer

to um to to this dynamic and i'm

i'm curious about whether that your

thoughts on that

in in the sense of um

you know do you do you see this as being
an a new
layer that's been added or is this
somehow
something which builds a bridge to the
way the structures could
emerge at the swarming um
is it because it talks about memory i
know yet the slide that talks about
memory i can't quite find which one it
is now but
one of the slides talks about memory is
that
significant in allowing the structuring
to happen
you know is it something about forming
that memory in that structure
or does the structure need to be there
first i'm sort of putting that out to us
and i don't know whether that's
something we could ask michael but
and maybe just refine that thought but
it seems to be
something which is a question that
relates to how this all fits together
nice dave
yeah for scott um if you want to see
how rhetoric is founded
in the basic emotions and also get an
evolutionary
biological evolutionary perspective on
that read the chapter on
play in yak poncep's
affective neuroscience
and once you've gotten through that look
at the later piaget's later work
on the role of play when he really got
very subtle about the socialization and

how
how kids uh use play to
become very expert in the in the ideal
case

in role taking and the creation of
uh uh social uh relations and
being able to play without anybody's
feelings getting hurt and so on
yeah the the concept of play i think is
really important
um in the space of active inference
because it really
kind of represents that that exploratory
component

scott
i'll give a thought while he stepped
away it was related to his
point about persuasion so if anything
it's not going to be a binary it's going
to be first off instrumental it's about
how it's being modeled we're not looking
for something that we're going to see
under a microscope or it
wouldn't even matter if we could we're
talking about categorization
so that we can reduce our uncertainty
about action
so the models that we set up could be a
binary state model or not
and that wouldn't be a metaphysical
claim but just thinking about it as
far as biological systems are concerned
and taking it to that sort of like
cellular
level like what if one molecule is
secreted
that sensitizes the cell to a second
molecule

you know is that type of sequencing
would that be considered a minimal
viable rhetoric or intracellular
narrative
and uh just made me think about how
there's sort of different types of
interactions
and that they convey different
information something that
my advisor professor gordon really
explored a lot in the ant colony
case how different types of interactions
just
update how likely an individual is to do
different kinds of actions
and that's seen across systems so
sometimes that's just a simple enough
statement like there's symbols that we
see you know
one or fifty or a thousand times a day
and then there's things that are more
explicitly
narrated more rarely so then in the case
of the human culture human narratives
it just opens up the design space
because there's multiple kinds of
interactions
nice scott did you want to give a
thought yeah a couple of things that is
really nice daniel that's like kind of
the notion
in introducing something like discretion
in a sense
so if a system is
running a let's let's call it active
inference
that running that model are they
are they bringing their model to bear on

the environment
because it embodies the information
of their past genetically and
epigenetically and behaviorally
so a three-legged dog is it running with
three legs just because now it has three
legs and that's it that's the affordance
or or is there discretion
in the active part of active inference
that may be an interesting dividing
point
in terms of rhetoric is rhetoric can
rhetoric be a persuasion
brought by and thing that's intrinsic so
a tiger jumps in front of me on a path
and i'm walking
the tiger is ferocious because it's an
apex predator
and has big teeth and i'm not is the
tiger
exercising discretion on being ferocious
or is it just
intrinsically ferocious to me
right so is so that's that to me that
feels like there may be something there
on rhetoric
if there's a choice maybe that the
organism has
or choices not maybe maybe the
organism's not a single
individual maybe it's a hive so
that's one thing then that gets to that
quorum sensing
which is really rhetorical argument
seems like it's because there there's a
choice shall we
swarm or not now when it's locusts and
it's a pheromone for gregariousness

right that's what switches the pheromone
switches on and off when they're in
close contact
so question is is that a discretionary
thing when there's a pheromone
impact you know again it doesn't have to
be in either or it's just that maybe
there's gradations of discretion that
are exercised
in different ways and so the other then
it gets to me
to get to synesthesia so
if i could sense something um i could
make a decision but
instead i'm fed a pheromone that makes
me feel
make a decision without my conscious
awareness
is that still rhetoric and is that a
quorum if it's a non-conscious
quorum maybe it's a different state when
it's non-conscious
and also when you have internal
synesthesia one of the things that's
fascinating to me i once heard that
the assertion that people with perfect
pitch might be synesthetes
because they're perceiving sound as
color for instance and they can discern
colors more readily in their mental
capacity
so therefore they can discern the notes
better than people who can't discern
notes as color
so again all those things feel like they
work together
in terms of is the consciousness that's
held is the mind

held in the individual brain of an
individual organism
and or in the communications among the
organisms
and so maybe that takes the rhetoric
question and renders it
a nullity because maybe it's degrees
of quorum that resemble discretion
and maybe the internal acts of
discretion are merely
priors that were learned from an
externality anyway anyway a few thoughts
yeah i'm really reminded scott and this
is going to turn you
up a lot this is going to turn up your
volume a lot
the uh so i just learned about the work
of um
asifa majeed i'm not sure if i'm saying
her name right but
she does this amazing work on language
how language shapes thought and how she
thought
shapes language um and she looks at at
like in cultures they have you know
there's was one i think it's the jahi
culture they have like
17 different kinds of smells that they
like smell and it's like you know one of
them is like
i mean we don't have words for smell we
just have it smells like a rose or it
smells like meat or
whatever so she you know but they have
actual in their vernacular
they have different words for different
smells and and
like in english we have you know a

gazillion words for color but like in
some cultures the color
blue and green are the same so so uh
and you know it's the words for like one
that was super amazing like we have all
this word for like fingertip fingernail
uh finger hand forearm bicep like
a shoulder i mean all of this in some
cultures is one word
your entire arm every part of it like so
you don't discern
that there's no difference between your
thumb and your shoulder like that's wild
to me right
and so how does that that shape the way
that we think about it uh sorry if i'm
rambling daniel
no fun points so as far as how
connectivity of communications related
to rhetorical
outcomes rhetorical ecosystems i think
there
an analogy to look for would be in the
connectivity of brain regions and
of course in the spm toolkit and the
dynamic causal modeling that we
referenced a bit earlier because there's
brain regions that are
anatomically connected so whether
they're next-door cells or whether
there's a long-range connection
and then there's also like functional
and effective connectivity which are
different ways of looking at the
signal of dynamical data and then
drawing edges
in a network where the nodes are the
regions and then the edges reflect like

a dynamical correlation or
granger causation which does of course
bring it into this whole
dynamic causal modeling the whole
question about multi-level
cause cause across different uh
parts of a system that aren't directly
touching is it just our physicalism and
our assumptions of materialism
that like require something to be
touching to have a direct influence
is it just some sort of anti-you know
non-locality
type stance that's very deep prior is
that something that when relaxed
provides useful solutions or just ones
that are
a bit less susceptible to one argument
but totally susceptible to other
arguments
nice questions very nice uh
stephen and then dave
yeah this idea like you the the question
about structure and the relationship
between
how things can relate or touch i think
it makes uh
the way that we bring in date our
sensorial information
and these cones in a way with a cell you
kind of
each cell presumably can have this
perception that can extend
we have these multiple threads of
perception that come in you know we've
got the visual perception we've got the
auditory perception we've got
and then how do we structure our

narratives

well if we have a culture which

structures the ability to

smell think about smell in that sort of

refined way compared to

us who maybe gets very involved in the

musculature and the nail and the

you know the amount of time people spend

now on their nails you know

in in in our culture so that will change

the structure of the narratives that

you're going to tell each other

and um that would be interesting and i

actually just did a laban

laban dance sort of workshop recently

and they have

they did the verticalities one of them

like you've got the base which is the uh

the feet and the floor you've got the

bowl of the pelvis that you can sort of

rock on you've got the diaphragm

you've got the base of the mouth and

you've got the top of the head as if

they're almost different

ways of grounding your ways of being

right and as you go up and down i think

that again

speaks to that structuring how much

would that change the structure of

the narrative and that would change the

structure of the memory

and how does that structure of memory

play into the structure and memory in

this

paper you know could be interesting

nice dave

yeah the uh different ways of

structuring language are

quite striking to me my um tagalog is
just dramatically different
in the way it expresses ideas from the
european languages
there's no gender for instance um
there are three cases nominative
genitive and
ablative and almost everything is
expressed in
all circumstances expects expressed in
the ablative
so where
when for what purpose by means of what
for whose benefit that's all just
expressed by putting
however many of those causes there are
into the ablative and if you need to be
specific
you throw particles in but they're
always optional
so uh if you challenge somebody well how
do you know whether you're
you know how what something means in
in a sentence so well isn't it obvious
if i talk about
monday that when you translate it to
english you'd say
on and when we're saying when i'm saying
at new york of course you would
translate that to
in and uh
for well of course you translate it
forth but when our before our son had
been in the u.s for a long time he was
constantly using this at
for everything at three o'clock at
manhattan
at um pittsburgh

at my girlfriend anything
other than who's doing the action but
who's doing the action isn't
expressed in a straightforward way
either so it's it's
it's pretty wacky
cool nice thoughts yeah
sorry pretty wacky is what anybody with
a highly precise language like russian
or chinese would say you know when they
hear somebody speaking even careful
english this is wait a minute you had a
not in your sense what is it exactly
that you're negating
there's no way to tell what an english
speaker means once he's put not into a
sense or in russian
it's so strict and so specific same
thing
you know how do you know in english well
you just figure it out
what don't you just say what you mean no
we don't
let you figure that out definitely some
ambiguity there i wonder like in cell to
cell communication
um if there's ambiguity like in in the
bioelectric
signal um like what which way does
should a cell grow or go
like touching back onto the the language
of this multi-scale cognition
daniel so i think to connect this uh
language discussion to the um like
the goal and the whole sensing point of
the paper
these different ways in which human
languages

are communicating state updates about
the world
people are able to live living people
use all these languages
so that says a lot it means they're able
to be uptaken
and transmitted by our cognitive
apparatus
and then also they're able to persist in
a niche
it doesn't mean that every language is
going to be as successful in
any niche but there is a place where
it's successful so just like
ants it's like that algorithms working
there
might be working somewhere else but it
might not be
so then we can look at this amazing
diversity of language and communication
structures including ones that don't
even
only use audio or ones that use audio
dimensions that we're not
you know using certain sounds or
something but then also the syntactic
and then like what is the invariance
what is actually being communicated
that's the structure of what is being
shared
between agents that are like one another
in certain dimensions
and they're sharing state updates so
that
they can make it happen in their niche
so it's like there's the total
the diversity is because there's enough
invariance for us to recognize it as

part of the same category
and then there's important things to
look at in what it is that's
in common
yeah definitely um it's definitely like
interesting to think about how
the the communication shapes
the structure shapes the all of that the
structure of society and the structure
of a human body
uh scott
you know it's so you guys are this is so
fantastic i
this is like everything
comes together in these conversations so
i was
i was thinking what is this reminding me
of i just read an article and i was like
is it this does it remind me of this
article which is about
universal chemical synthesis computing
no
was it about this highly multiplex
spatial mapping
of microbial communities sort of but no
what this reminds me of is this space
time from bits
this guy i just read this article it's
about quantum gravity
and what he asserts is gravity is what
we perceive
as it's actually entanglement of bits of
information
and so if you take out the entanglement
you have no space time
and what i'm thinking is the
entanglements we just described
that's everything we call it mind we

call it society we call it eusociality
we call it space-time
it's edges and we're characterizing
these edges
and our affordances are so constrained
that we try to describe it but we're
describing it
it's the shadows in the cave right we're
seeing in
just indirectly what's actually out
there and one of the things that's
enriching about this active inference
notion
is it provides us with a framing that's
sufficiently
generic but also sufficiently robust
that it explains things in the stack and
the time stack and the space stack
it's kind of like wow this may be how
things interact
maybe things are entangled and
we use the word internal model and
external reality
but maybe the nature of rhetoric and
maybe the nature of entanglement
is the seeking of connection
by neg entropy systems so i you know
my life definition is um that
living systems are auto catalytic and
entropy secreting
my son said i just wanted to use the
word secreting because it sounded gushy
but if they're auto catalytic and
entropy secreting then if you're a
system that's living you're going to
want to seek out other living systems
because you need reproductive and
nutritional opportunities

in order to be auto catalytic and
entropy secreting
and you know it goes to the frog nervous
system rules right if you if frog
encounters something that's smaller than
it and it moves it eats it if it's
larger than it and it moves it runs away
and if it's the same size as it and it
moves it mates with it
very simple rules that let a frog be a
frog
right so it feels like maybe what we're
banging up against
is too much too many words that we're
throwing at
simple concept and maybe we're talking
about is harmonic
coupling of living systems that are
seeking other neg entropy states
either to consume or mate with
the last point i have a poster that i
got from a guy in san francisco
and it's in my shop and it says futurism
imagining a fake future in the hope that
the real future will come along and mate
with it
and that's kind of like what we're we do
in active inference we're imagining
our way into a we're perceiving an
externality just because our perception
say okay there's something you don't
control
out there and then we try to mate with
it or eat it or run away from it
i was reminded of the light cone slide
here when we're talking about space and
time and edges and
um how that daniel included the slide

how
like light is is the boundary between
space and time
like that's kind of that trips me out um
steven and then daniel
yeah i think when we be interesting to
also i think this
opens up maybe a revisiting or with this
with chris fields and some of this work
about
quantum contextuality as well because as
you start to
because one thing he did mention in his
presentation you know
there's this question about when it goes
from being contextual which is
or quantum like which is a highly
constrained
set of options basically
once you've got more options for
movement things start to move into
more classical even if there's loads of
quantum stuff going on at the small
scale they all average out and
effectively
the gas in my room is behaving similar
to the gases in all the rooms that we're
in
in a general generic sense only if we
went into the sort of and and there's a
question that comes up then around this
sequence in
um and um because
where where is that sequencing possibly
constrained enough
that you might hit some things like that
and is that part of what happens
at these well you could call them

meso levels i mean the cell the surface
of a
cell is small but it's not small from
an atomic point of view it's still
relatively aggregated
yet we're getting these distinct flips
in
charge and is it plausible that anything
as
big as the brain can also do something
like that
and it may be that the term quantum is
not necessary that in that sense but
um yeah it'd be interesting to think
about what it what what
importance is there to be a constrained
set of sequences a bit like daniel
mentioned if
a certain molecule is released and then
another one's released and then another
one's released
that could be some sort of type of
rhetorical dynamic but that
there needs to be a constraint so that
it's not just part of the
classical noise you know it's not just
swamped by
a big flux of um randomness right that
it actually is a sequence
even if it's not as clear as you know
so that might be an interesting question
i don't know whether it's a fair
question to put
to mike if that's is or maybe it's more
of a chris fields type question but i
know they work together so
they might have talked about some of
this

definitely daniel and then scott
yep it was a key theme in this paper
and in cybernetics in general that like
goal seeking and
planning as inference is a big area
that's why we're always talking about
like control theory
and it is interesting to wonder how does
the bioelectricity play a role in
goal setting and then in basically
policy selection because you know active
inference we don't
just specify which trajectory is going
to be like the preferred one and then
condense it down to a single summary
statistic it's like
there's the preferred states with the c
and then the affordances with the e
and then there's a equation that is
selecting policies
based upon you know a combination of
different ways of phrasing some of the
variables
so then it's like how does that framing
which is not just one way of
going about doing what is it gonna allow
us to say about
like how systems actually plan
i think that's kind of related to the
hint in figure 2
on 32 like possible alien come
ai it's saying that this framework
wouldn't just help us describe
systems but recognize systems that we
didn't
have ways of recognizing otherwise
perhaps
nice point and if um you know if another

system

was communicating with us how would we
know right like what what would that
even look like scott

i was um have you guys seen this article
decoding and perturbing decision states
in real time now

it's um p-e-i-x-o-t-o diogo

pixoto it's it's about decision states
it seems like it's very similar to what
we're talking about here

um 25th of march 2021 from nature
magazine

um and then bearheim v-e-r-h-e-i-n is
another author

um one of the things the reason the
reason i was holding that up
one of the things i was just writing
down you know light cones

lights um bosons right

photons are bot or bosons and one of the
things i always fuss with a little bit
is

okay is there a difference between
fermionic and bosonic

physics here that we need to attend to
so the idea

of fermions i always think of them as
firm so they make up matter and bosons
make up energy stuff

and information is carried by bosons but
also carried by embodiments in
in fermions and stuff and so

it's so interesting to me that how does
the

persuasion the rhetoric

the the history

get embodied both both sonically and

fermionically again you have the peacock
and you have the birds and the and the
animals that develop certain
physicalities over time as communication
right it and it's a deep time
communication and there's less
discretion in it let's say they can
exercise it or not but they have the
embodi it's embodied in the physicality
and then there's the
action that they take which is
again they have the physicality with
which they act
so what are the instruments given them
physically but then they
convey information by the behaviors they
engage in and i keep coming back and
thinking
well it's kind of interesting if we look
at the carrier signal
again of information um
is it in fact bosonic always it's
things are carried by light i guess
things carried by sound are being
carried fermionically by the
pressure in the sound wave i'm not sure
but
so maybe the medium i was just wondering
whether the medium
relates to the notion of rhetoric and if
there's a pure
form of information transfer when it's
bosonic rather than fermionic and maybe
that's getting
way too reductionist on the thing but it
i wondered whether
light cones in their purest form which
presumably are

involved light right the original light
traveling out and it can only go this
way and you can't be beyond it
and blue when you said before it was a
statement
that what is it outside the light cone
there is no
it's like the defines the edge of space
time for the observer
and so what's what's out there is an
externality
it feels like that boundary
when that's known to other parties maybe
in communication we're trying to do
is think outside the light cone
i don't know
uh daniel and then stephen so the
part of the light cone going forward in
time is anticipation
and if we take anticipation again not to
be like a metaphysical claim about what
systems are really doing
but what the system is anticipating
outwards
that entails a generative model through
time
of the states that matter for this kind
of an agent which is
the internal state of the model itself
the blanket states which are sense in
action
expected outcomes and expected actions
and then the state of the niche the
external states
so kind of operationalizing the light
cone as a specific
prediction on specific variables
might be helpful and kind of give a

little bit more insight into how it
could be used
in a computer and then also just to
scout your point about like the media's
the message essentially
it's like a multi-scale media is the
message it's not just
um bold print type versus cursive
it's actually like you know you're
bringing up types of subatomic particles
like that media you know is ether the
ultimate media
like the nature of the media does matter
and yeah
it's a trade-off space like with
peptides
you have a larger signaling space than
with small molecules like the
combinatorics are just different
but that entails totally new challenges
like with specificity or
cross-activation of neuropeptides or
how do you keep them linked through
evolutionary time and then we see sort
of
insights into those design trade-offs
when we see like
wow there are like a lot of
neuropeptides that come from one
precursor
longer protein so that suggests that
their co-regulation is held together
through specific design principles so
those are really interesting
questions about how like the speed and
the frequency and the bandwidth
of different information channels
structure

how they end up performing
nice stephen and then scott
yeah i think daniel you're making a good
point and it feeds on
with um this question about time scales
and
is if we i assume that slides up the one
that's there with the tick and the dog
and
human and possible alien um and
so the cones there yes it's very useful
to think of it
in terms of i've got my maximum
information now maybe my maximum
information was even a little bit in the
past
but it depends on what regime of
attention i was applying right i
i made choices so as i make choices into
the present
i reduce the possible cone
extent i've got available you know um
and
as i bring things only into my conscious
awareness you know um
certain things are going to be left out
and then bringing that
into okay this is a particular type of
future i might predict
based on my regime of attention is
actually probably
more useful in some ways
to this because the cone going out it's
a nice idea but it can lead to thinking
that we know more and more in the future
there's more more futures out there
which is there is more more futures but
not necessarily coming out from us

um so it could be this is a bit more
like okay my cone could move around a
lot
and there is actually a lot of analogy
with that because now a lot of
futures work they're actually saying
you're better off picking a scenario in
the future
building out the scenario in the future
as a separate thing
not inheriting your present because that
will always bias you
and then bring it back bring your
present which is what you
know and your assumptions here and sort
of bridging the two
in terms of the narrative rather than
this idea of i can
go because i'm could just be going with
all my
assumptions and all my pre-assumptions
from the present
and making all these stories about the
future um
and um and starting it almost starts
from a singularity in those other
diagrams it's like goes from a
singularity and opens up whereas
actually really
the widest is now potentially this is
the most information i have potentially
available
or at least within the 10 or 20 seconds
of this moment
um so i think that actually
there's other ways that this could
change the way people think about
sense making in time and space

really super nice point stephen um scott
wow stephen wow i mean that's so
awesome when it makes me think okay
first of all
encoding and decoding okay so something
i i just wrote down mutations
encode new speculative space time
so mutation gives you access it's a
random mutation
that gives you access to space time you
weren't intending to access
and the same thing with externalities if
another thing comes in then this is jazz
this is jazz genetics right is you get
mutations and it's like whoa i didn't
want to access that
that new like cone area but i guess i'm
there now
okay let's see what we got here and
that's pretty awesome
because otherwise you'd be settled in
your ways and you'd never see that space
that this flight cone area would never
open up
and you don't know what it's going to be
you don't know it's like it's
i always said to my kids life is a
one-act play no rehearsals
that's a light cone right you just go
through and it's like okay here we go
so that idea of encoding
and decoding is kind of interesting so
you have this the
we we go out there in the world and
we're like okay i want to interpret the
world
now some things were encoded with
intention to communicate

music speech some things were encoded
with intention to observe right
cryptography stuff whatever
some things were encoded with no
intention to communicate but they do
communicate like
oh leaves are green so therefore over
time we figure out that they have
magnesium in the photosynthetic cycle or
whatever right so they're not
the color is encoded in the thing but
the tree is not intending to say i'm
doing photosynthesis right it's just
we now take that perception
and can decode something that was never
encoded with
discretion i guess or intention by the
organism or whatever you know we don't
run going with that
so this notion of these light cones
and it from the perceiver's perspective
we can decode within a light cone we can
decide an
artist decodes a light cone and somebody
else seeing the same perceptions would
not have the same light cone
right somebody with different priors can
see things a person who knows history
will not repeat it because they're like
they see the same history now and they
go
whoa whoa whoa whoa that looks like an
insurrection
like their light cone is more trained to
the possibility that's an insurrection
so this last couple of comments are
fascinating to me
that we're moving through space we have

this active inference model
we're running a model there's myriad
other models and non-models out there
that we're seeking to decode
some subset of them were intended to be
coding information
some subset or not and we're just moving
through space decoding everything
some of it will be false signal some of
the actual signal
and so the light cones the experience we
have
obviously can never be replicated by
another person with different priors
but also in real time
it's a i don't know what the equivalent
is of a cacophony of light cones
a babble of light cones or it's like
that ice cream cone again there's too
much ice cream maybe in the cone
anyway it's fascinating notion of that
over
decoding in a way and what and maybe the
scientific method
has something to do with a rigor on the
over decoding that can happen with
alchemy or something like that
nice point daniel and then stephen
so to scott's point about the mutation
as uh
accessing spatial temporal uh
zones like expanding uh into space-time
i thought okay maybe there's some like
quantum you know 4d
three plus one interpretation of that
but
how can we think about that with the
information that we already empirically

know

and it's actually that mutations in

developmental biology

developmental genetics facilitate the

exploration

of new developmental space and time like

by slowing down or speeding up the way

that a certain cell type

is uh susceptible to a hormone or

something like that

so what's being played with is actually

the level faster

because that's why there's the

intergenerational transmission of those

mutations

but then what they're actually modifying

like the source code but

not exactly in that way of course is how

the system bootstraps and modifies

itself through development and learning

and then how can we think about active

inference in that way

well we can think about the mutations as

modifying like the initial parameters or

some of the

state variables for the parameters that

are important for an active inference

agent like

the structure of its internal generative

model its blanket states like its

sensory and action affordances and then

the niche

so that's a way where we can have the

generality to talk about like

potentially a system across different

scales

but then when we're ready to actually

formalize it and then

ask how to design communication in that niche
nice point stephen
yeah that's a good point thinking about those different time scales and and sort of following on from that you've got this question of the um so maybe the things are happening at this light speed and there's a lot of electrical activity happening in these cells and groups of cells and and we've got these structures that we can come together and of course we then have this intermodal integration between different cells nerves coming in and the question then also is as it's coming in say from my body to my brain and my feet are seeing the world in the past compared to my nose right because it takes longer for the sort of information to travel up from my foot the signals at the speed of light traveling but by the time it does that through all the chemical interactions it you know it's milliseconds not billionths of a second so it's coming together in this structure of the body which is grown to you know to integrate it but at some point the system has to calibrate

it has to calibrate so that it doesn't think that my foot is in a different time to my nose or my ears or my fingers right so that sort of brings that question of of um you know some of this multi-scale stuff will come out of these kind of electrical light speed dynamics but then it's going to also come into the uh the structures i think the daniel was talking about structures we all sort of mentioned structures and then there's going to be a system that sort of at times has to sort of tweak it you know and maybe that system's not needed when you've got a small set of cells that are all firing electrically and doing other chemical signals they just the structure's enough in a way but uh um so yeah i thought that that might be kind of that might be an interesting question to ask michael as well if this you know how does how does that kind of play out with this nice daniel and then scott yeah just a quick comment on that so the signal processing or the sort of outside in perspective would suggest that there would have to be like a recalibration then the generative model out way of thinking is that the generative model

is that like your hands clap and the
sound and the visual
are the same event so there isn't as
much of a calibration of
inputs as there is a reconciliation
of deep temporal models which are timed
based upon
the inferred causal structure of the
world which is usually not two separate
events
but rather one and then that is actually
just
embodied in the sensory relay which does
take different amounts of time
to transfer from different parts of the
body
nice scott yeah okay so i just was just
one thing on that i'm just before scott
that's a good point i think
that's interesting you know thinking
about predictive processing
and not getting caught up in the in
that's where it's tricky because it's
it's not directly an input it's a kind
of
error message so yeah that's cool that's
not the part that's experienced
yeah and the parsing of that opens up
awareness right if you can parse the
different
pieces and start to break it down this
last couple of pieces i
thought you know the occupy movement
occupy wall street
it feels like active inference we should
have t-shirts made up occupy light cone
because everyone's every organism is
trying to occupy everyone else's light

cone

i think i mean and when i so we worked

for microsoft for

at the law school we had a program

microsoft paid

professors intellectual property

professors to go to china

to teach intellectual property

professors in china and in india

usip law concepts

now why why was microsoft doing that

they were doing that because they were

trying to render

the externality innocuous there was a

piracy externality

that they were dealing with and so they

thought if they teach

the teachers in those jurisdictions

about the value of intellectual property

and then teach the lawyers at the law

schools

the teachers teach the lawyers those

notions then the microsoft notions of

intellectual property

they'll be left software privacy so that

was they weren't doing it for

beneficence they were doing it to render

the externality innocuous

they were trying to occupy the light

cone

of a bit of a culture right with this

affordance

uh shifting by mutating the

teachers views and the way they taught

intellectual property law

so that that to me

there's a i once heard a quote that

trees were bullies because trees suck up

the water and the light
in an area right you know they bully
other plants and i always thought you
know trees are so nice and i always
think of trees as bullies now
and it's kind of interesting when some
when an organism
is applying rhetoric or other vehicles
to sweep in and
you know like pigeons when they do
pigeon racing when you try to pigeons
try to
flock and you try to grab everyone
else's pigeons right
or i don't know in hive behaviors if
hives pick up other bees
and that makes them fitter as hives or
if there's mixing of bees when they do
swarming i don't know the answer to that
i just don't know enough about bee
biology
but it is kind of interesting that light
cones
maybe are pretty darn aggressive for
living things
trying to render externalities innocuous
and maybe that goes to gaia theory
that ultimately what what's the light
cone for gaia
yeah the biggest light cone of them all
just what you were saying made me think
of maybe
you know we think about it as a light
cone but maybe what we're looking at is
like an information cone you know and so
like
like a concentration of information
within an individual because in the end

we dissipate
you know there's no more information
concentrated there uh stephen
yeah and it might be a question when we
think about this term light cone
and i was walking home last night and
i could see the clouds and i could see
the lightning
but i could only hear the thunder you
know five seconds late or whatever so
the information cone on the audio was
different to the information
if it is an information cone and then i
don't know the smell of the
the rain or the change of that so um
i might be interested in how much the
the
there may be some elements in certain
scales where really it's light
it's about the speed of light i mean
near enough so that's what it is or
maybe it's a broader category which
could roll
out into these other spatial temporal
modalities
yeah the uh so i mean you know when
you're in a space if you wake up in a
dark room what's the first thing you do
like where am i right turn on the light
that's a very human um way to maximize
your information gain
i mean not every organism obviously can
see like uh daniel
i think my closing thought on this would
just be that
it really speaks steven's story about
how there actually was a temporal
structuring even though you cognitively

knew it was a single event
you actually classified it into a
separate kind of event which is like the
exception that proves the rule
that suggested long long ago that the
speed of sound and light were
different how are we going to make sense
of this total continuum of
signaling in the niche we can think
about ostensive cues and helping you
plan

there's something when you smell
something and you have eight hours to
plan for the storm versus you know you
hear the lightning strike you have a
half second to plan

et cetera so we can think about the
structuring of different kinds of cues
in the niche as giving information that
is

in light of a generative model able to
be used for anticipation in action
so that's just speaking to how
core active inferences to thinking about
these different processes coming
together

and that's why it'll be awesome to get
the author's perspective
on all of this and what active inference
is going to do

nice scott you have a final thought for
us

yeah just again that i'm still intrigued
by the notion of the entanglement
creating space time and this whole
discussion of
time cones and the time cones in the
relation of space-time

relativity theory etc and uh i feel like
this

one thing that active inference has done
whatever it's and there's so it's
powerful in many ways

one thing it's done is allowed me to
open up my mind to processes
that feel like they're replicated
across time and scales

and that's interesting because whether
or not they're strictly
fall within active inference math and
rigor

the the model has
uh opened up so many different ways of
thinking about things

and the entanglement notion where we
think of two

is it two particles or one spooky action
at a distance

it feels like everything else we're
talking about active inference is spooky
action at a distance for me

it's just that it's a different spooky
action at a distance and so

it's fascinating the notion of
entanglements and it feels like light
cones

may be describing the things that are
amenable to in

be entangled or coupled and so maybe
if it's an information cone that's an
interesting

notion that we can start to say
communities expand out

that which is amenable to coupling
because they widen

the light cones so that using that

rhetoric

may start to be a way that we can

illustrate the benefits

of community-based behaviors

and things like that so we can start to

bring this back and say look

you can't see around the corner but that

person who's up there they're standing

there and they can see around the corner

and if they can talk to you they can

tell you what's around the corner

so maybe that's good right and so maybe

that's contact tracing for

pandemics maybe that's you know giving

the neighborhood watch for

other toxic things in the neighborhood

environment whatever

so it feels like this gets simple enough

you know people may not want to know

about light cones or whatever but they

it starts to get the entanglement and

that notion of

cause and effect feels like there's a

story that we can tell here

and illustrate pretty simply

anyway it's exciting it's an exciting

conversation some wonderful stuff and

it's delightful to be back in the

conversation

so just uh to come to the author's like

final point

in this um you know the dissolution

of the self right this is like the the

practice of of

zen um he says it is striking that the

process which

zen practice is meant to reverse

attachment to past memories and high

valence for future expectations or fears
is precisely the process suggested to be
responsible for the creation of complex
selves

so i want to know what happens to the
informational

light cone in this like dissolution of
the self does it like you know turn into
lots of little light cones as is kind of
implied in like the

the one figure but or does it just like
poof and like dissipate i don't know
it's it's interesting to think about

um that for me he says it's unclear
whether it's beneficial or even possible
to truly live in the moment and let go
of past memories and future expectations
but anyone who succeeded in doing this
would achieve precisely what zen
promises

the dissolution of the self um so thanks
everyone this has been a super
interesting discussion

and i am looking forward to our
conversation hopefully with the author
next week thanks nice work blue
thanks thanks very much take care have a
great day bye everyone